Beyond the Prompt: Leveraging Large Language Models (LLMs) for MS Care and Research

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10th Joint ACTRIMS-ECTRIMS Meeting

OCTOBER 21-23, 2026 | TORONTO, CANADA

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Possible LLM workflow



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*Image generated with Copilot

Possible LLM workflow



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Educational objectives

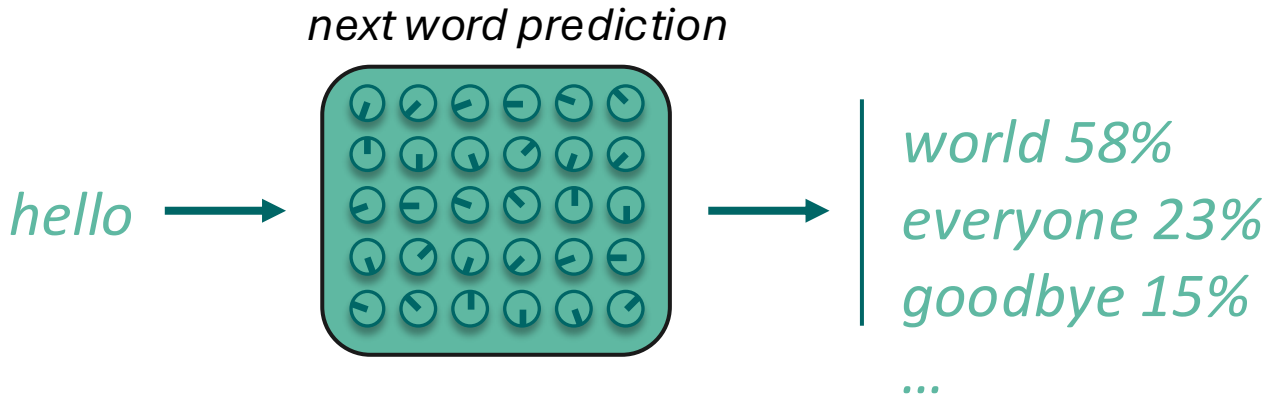
After this talk, you will be able to answer:

- What are LLMs and how do they work?
- What role can LLMs play in MS care and research?
- How can I safely and effectively use LLMs?

What are LLMs and how do they work?

Section 1

What are LLMs?





- Understanding
- Generation

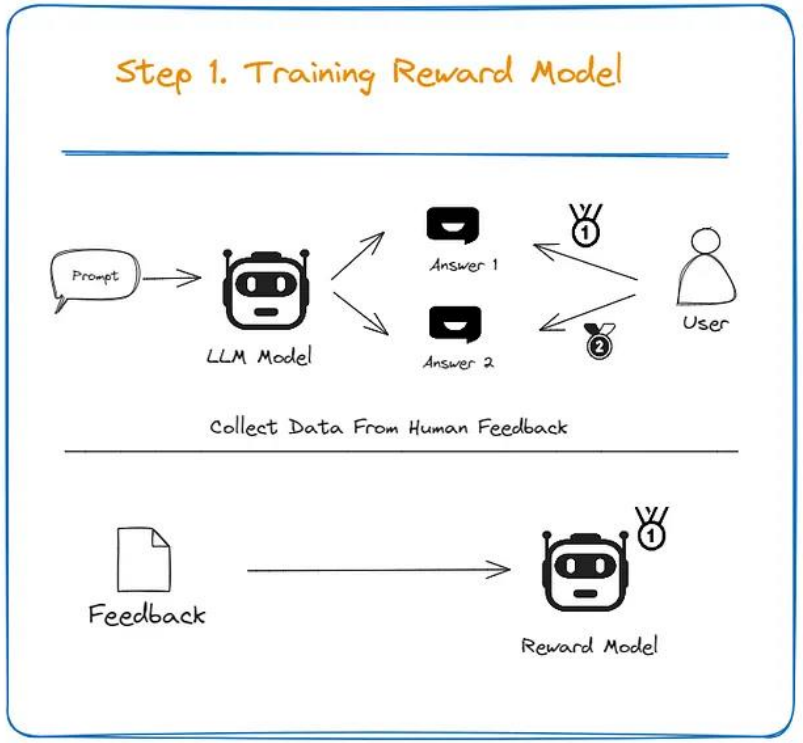
- Transformer architecture
  - Contextual nuance and word relationships
  - Key limitation: No built-in “truth”
  - *Base LLM*

Dijkstra et al., 2025 & Yu et al., 2023

How to improve LLM performance?

From base to fine-tuned model

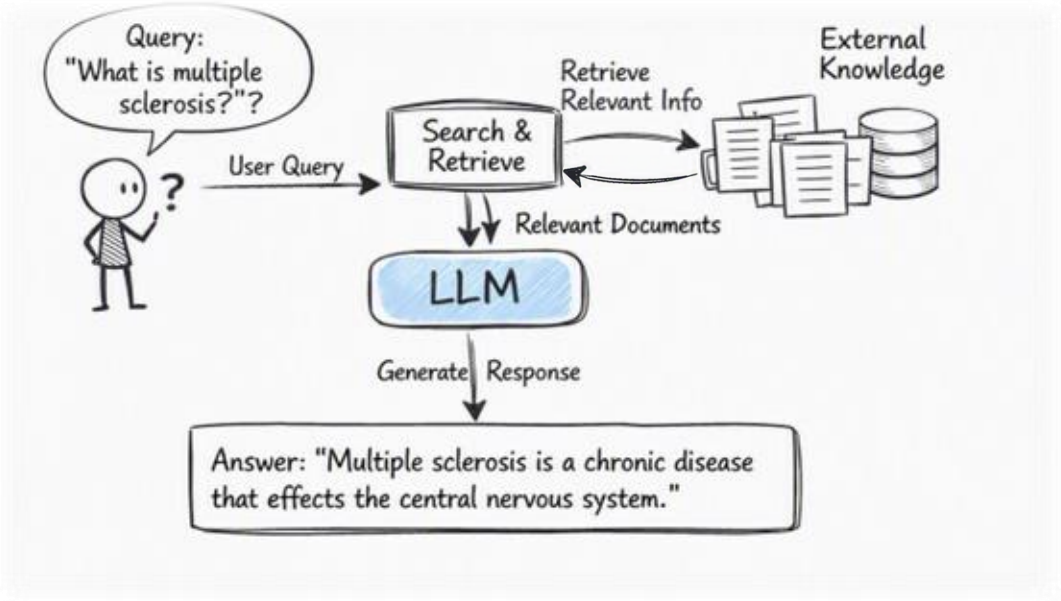
Reinforcement Learning from Human Feedback



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Domain grounding

Retrieval-Augmented Generation (RAG)



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ºImage created by Luc Nguyen
*Image generated with Copilot

Yu et al., 2023 & Inojosa et al., 2024

What role can LLMs play in MS care and research?

Section 2

Why LLMs in healthcare?

- Many stakeholders
  → different communication styles





- An information-rich field
  → need to synthesize efficiently





- Time is limited
  → need for efficiency

Yu et al., 2023

LLMs in healthcare

Recommended reads



communications medicine

PERSPECTIVE

https://doi.org/10.1038/s43856-023-00370-1 OPEN

The future landscape of large language models in medicine

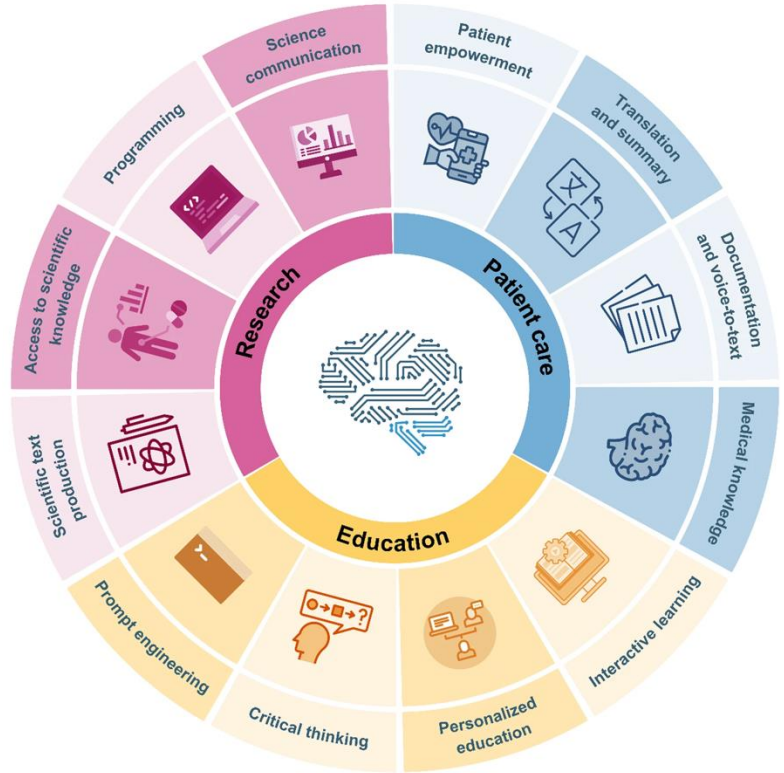
Jan Clusmann[1,2,8], Fiona R. Kolbinger[1,3,8], Hannah Sophie Muti[1,3,8], Zunamys I. Carrero[1], Jan-Niklas Eckardt[1,4], Narmin Ghaffari Laleh[1,2], Chiara Maria Lavinia Löffler[1,4], Sophie-Caroline Schwarzkopf[3], Michaela Unger[1], Gregory P. Veldhuizen[1], Sophia J. Wagner[5,6] & Jakob Nikolas Kather[1,2,4,7]✉

MULTIPLE SCLEROSIS JOURNAL | MSJ

Topical Review

Integrating large language models in care, research, and education in multiple sclerosis management

Hernan Inojosa, Isabel Voigt, Judith Wenk, Dyke Ferber, Isabella Wiest, Dario Antweiler, Eva Weicken, Stephen Gilbert, Jakob Nikolas Kather, Katja Akgün and Tjalf Ziemssen



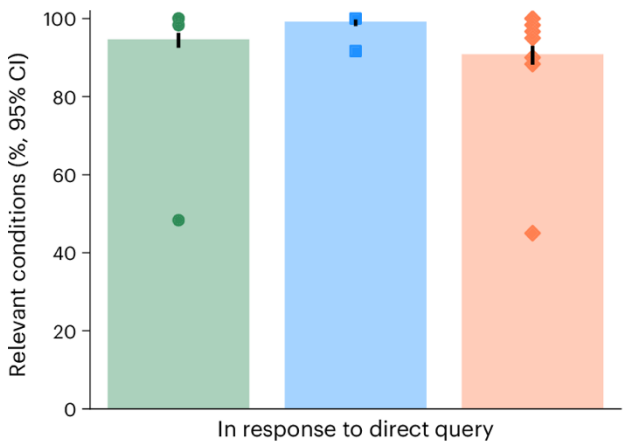
Clusmann et al., 2023 & Inojosa et al., 2024

LLMs for MS

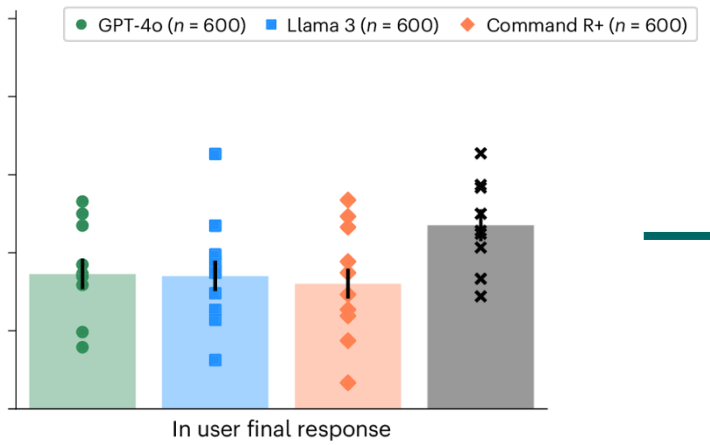
Considerations for patient care

Patient empowerment

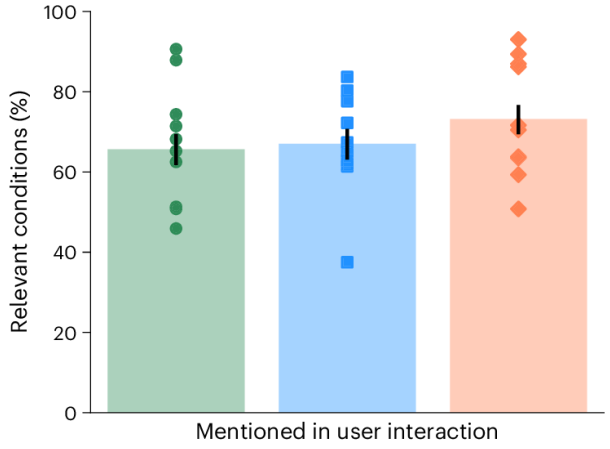
- MS patient adherence can impact treatment success
  - → Potential: instant responses when professional is unavailable
  - → Risk: incorrect conclusions by user



LLM <u>without</u> user



LLM <u>with</u> user



Inojosa et al., 2024 & Bean et al., 2026

LLMs for MS

Considerations for patient care

“I feel more tired during summer season, what shall I do?“

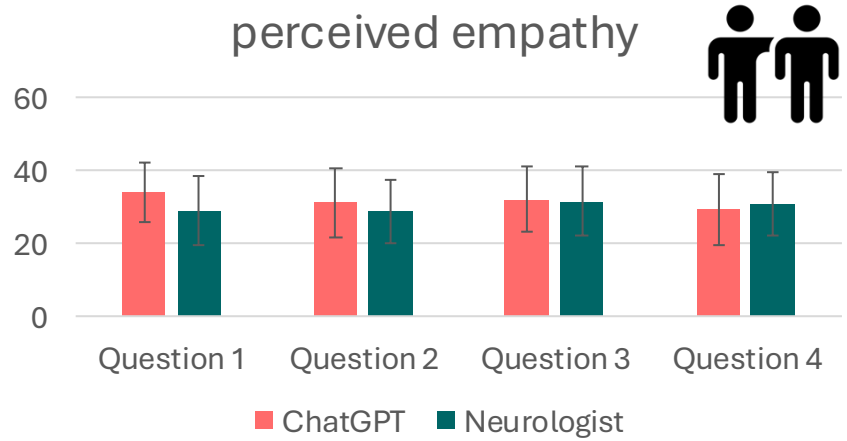


Translation

- More digestible dialogues
  → Uncertain prognosis

- ChatGPT perceived as more empathetic
  - Confidence and informality?
  - No differences in satisfaction

- LLM as valuable support tool



Inojosa et al., 2024 & Maida et al., 2024

LLMs for MS

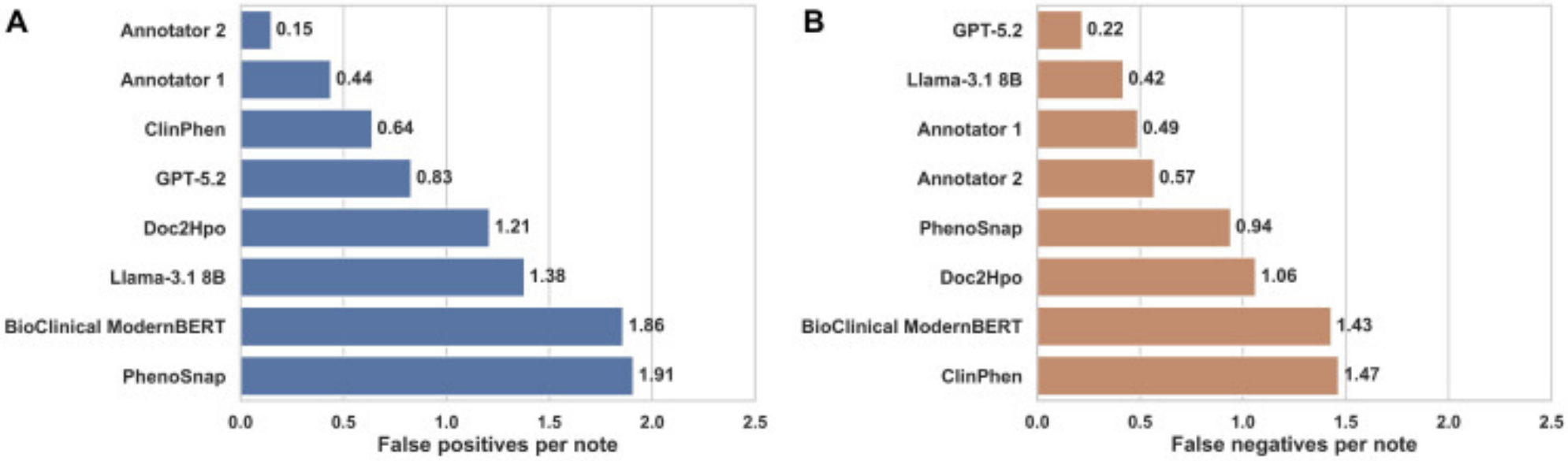
Considerations for research

Phenotyping from clinical notes



Access to scientific knowledge

- Help digest real-world data



Hier et al., 2026

How can I safely and effectively use LLMs?

Section 3

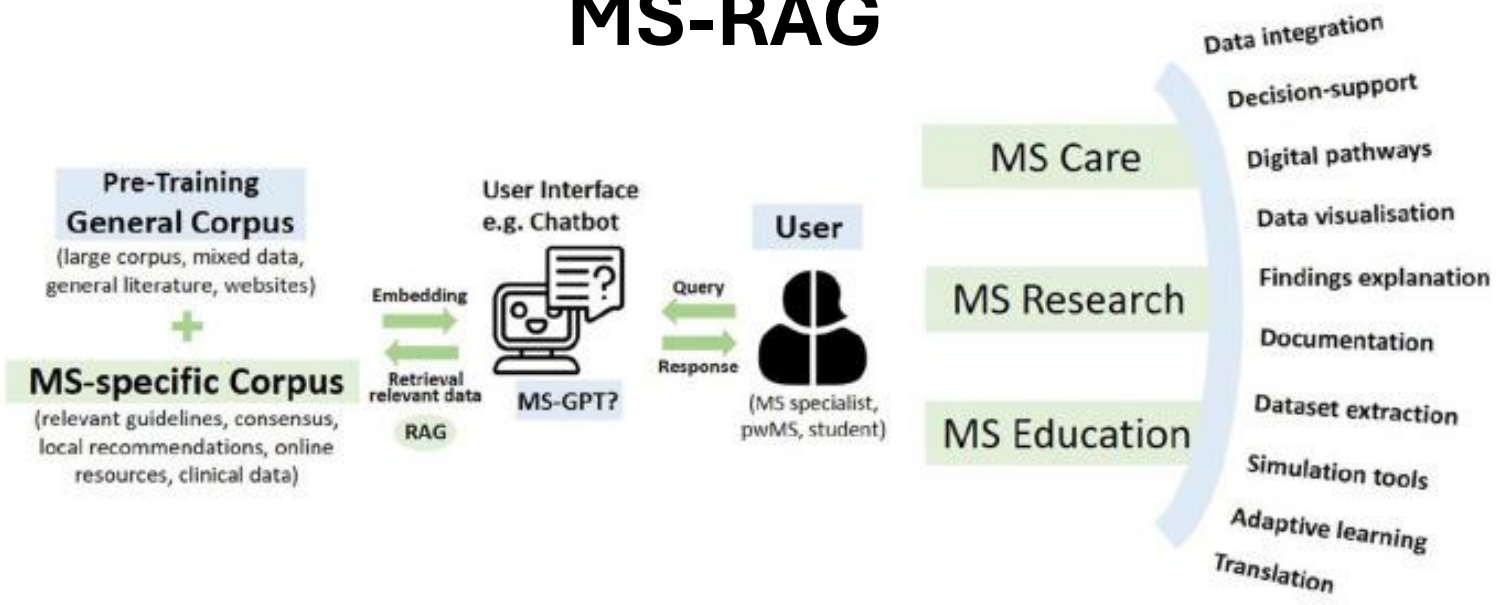
Safe use

Be careful with patient data
→Speak to your institutional expert

Hallucination

- Answer sounds plausible but is unverified or false
  → Treat output as draft

MS-RAG



Bias

- Answer reflects skew in training data

→Validate diversely & maintain human control

Yu et al., 2023 & Inojosa et al., 2024

Mindful use

Environmental impact

- *Little transparency*
- *Small language models*

Human susceptibility

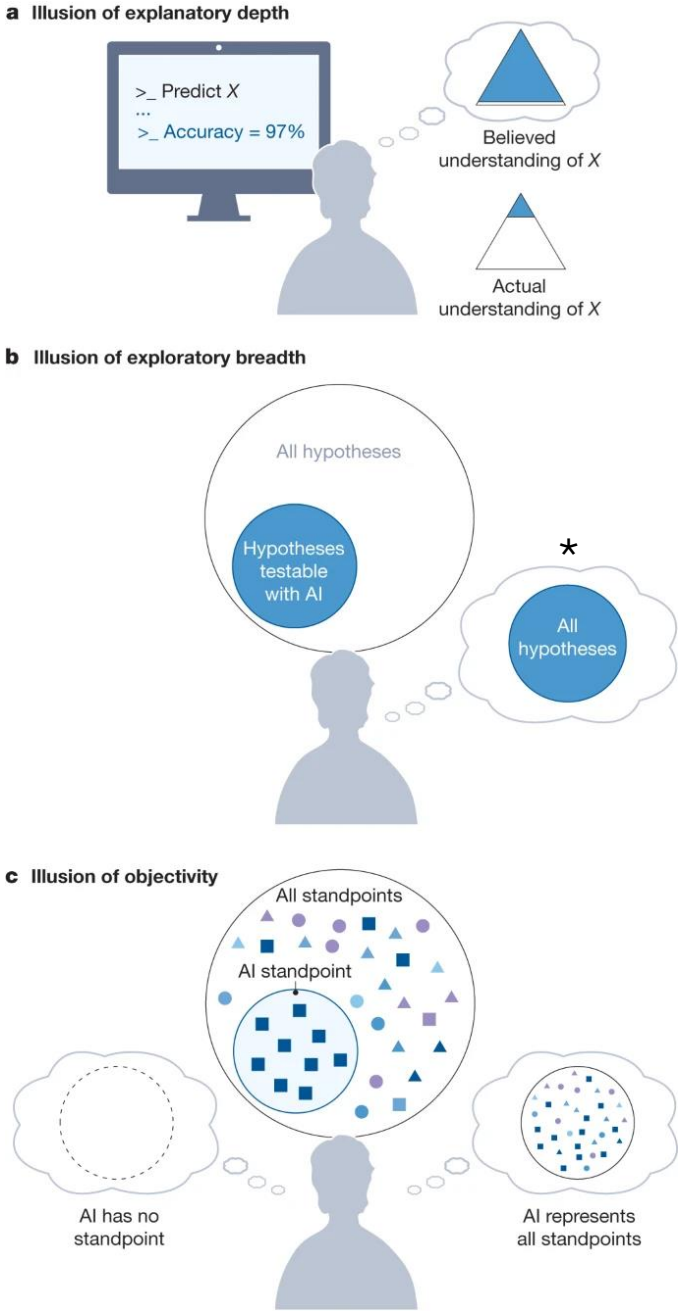
- Automation bias
- Illusions of …
  - *Explanatory depth*
  - *Exploratory breadth*
  - *Objectivity*



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AUTOMATION BIAS



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ºImage created by Akanksha Sinha
¤Image retrieved from cloud-science.de

Ren et al., 2024 & Juanquarter et al., 2026
Qazi et al., 2026 & Messeri et al., 2024

Effective use

Prompt engineering

- Model-dependent
  - Changes rapidly

General advice:



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*Image generated with Copilot

Meskó et al., 2023

We need you to shape how AI is used in MS!

Wrap-up

iAIMS
international Community
for AI in Multiple Sclerosis

Thank you



Join us:
https://iaims.org/

Amsterdam UMC

MS Centrum Amsterdam
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Amsterdam
Clinical Neuroscience

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References and resources

References

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- Icons: https://thenounproject.com/